

TUGAS PRAKTIK KOMPREHENSIF
PRAKTIKUM 1 – SISTEM OPERASI



Dosen pengampu
Triawan Adi Cahyanto M.Kom

Disusun Oleh :
Divo abdil arkananta (2410651059)

Tugas 1: Process Manager

```
divo@MSI: ~  
process_manager.c:118:9: error: expected declaration or statement at end of input  
divo@MSI:~$ nano process_manager.c  
divo@MSI:~$ error: expected ';' at end of input  
error: command not found  
^: command not found  
divo@MSI:~$ nano process_manager.c  
divo@MSI:~$ nano process_manager.c  
divo@MSI:~$ gcc -o process_manager process_manager.c  
divo@MSI:~$ ./process_manager  
Child1 - Masukkan angka faktorial: Child2 - Masukkan batas bilangan prima: Child3 - Masukkan string: ^C  
SIGINT diterima, menghentikan semua child...  
  
Child 1 (PID 5958) selesai  
Hasil:  
Exit status: -1  
Waktu eksekusi: 9066 ms  
  
Child 2 (PID 5959) selesai  
Hasil:  
Exit status: -1  
Waktu eksekusi: 0 ms  
  
Child 3 (PID 5960) selesai  
Hasil:  
Exit status: -1  
Waktu eksekusi: 0 ms  
divo@MSI:~$ ./process_manager  
Child1 - Masukkan angka faktorial: Child2 - Masukkan batas bilangan prima: Child3 - Masukkan string: 5  
  
Child 1 (PID 5963) selesai  
Hasil: 120  
Exit status: 0  
Waktu eksekusi: 17737 ms  
^C  
  
SIGINT diterima, menghentikan semua child...  
SIGINT diterima, menghentikan semua child...  
  
SIGINT diterima, menghentikan semua child...  
  
Child 2 (PID 5964) selesai  
Hasil:  
Exit status: -1  
Waktu eksekusi: 129726 ms  
  
Child 3 (PID 5965) selesai  
Hasil:  
Exit status: -1  
Waktu eksekusi: 0 ms  
divo@MSI:~$ nano file_monitor.c  
divo@MSI:~$ #include <stdio.h>
```

Tugas 2: File Monitor

```
divo@MSI:~  
Hasil:  
Exit status: -1  
Waktu eksekusi: 129726 ms  
  
Child 3 (PID 5965) selesai  
Hasil:  
Exit status: -1  
Waktu eksekusi: 0 ms  
divo@MSI:~$ nano file_monitor.c  
divo@MSI:~$ #include <stdio.h>  
divo@MSI:~$ #include <unistd.h>  
divo@MSI:~$ #include <sys/types.h>  
divo@MSI:~$ #include <sys/stat.h>  
divo@MSI:~$ #include <fcntl.h>  
divo@MSI:~$ #include <sys/time.h>  
divo@MSI:~$ #include <sys/wait.h>  
divo@MSI:~$ gcc -o file_monitor file_monitor.c  
divo@MSI:~$ ./file_monitor  
Masukkan direktori untuk dipantau: 9  
inotify_add_watch: No such file or directory  
divo@MSI:~$ nano file_monitor.c  
divo@MSI:~$ ./file_monitor  
Masukkan direktori untuk dipantau: 67  
inotify_add_watch: No such file or directory  
divo@MSI:~$ gcc -o file_monitor file_monitor.c  
divo@MSI:~$ nano file_monitor.c  
divo@MSI:~$ #include <stdio.h>  
divo@MSI:~$ #include <unistd.h>  
divo@MSI:~$ #include <sys/types.h>  
divo@MSI:~$ #include <sys/stat.h>  
divo@MSI:~$ #include <fcntl.h>  
divo@MSI:~$ #include <sys/time.h>  
divo@MSI:~$ #include <sys/wait.h>  
divo@MSI:~$ gcc -o file_monitor file_monitor.c  
divo@MSI:~$ ./file_monitor  
Masukkan NIM: 9  
Masukkan path direktori untuk dipantau: ^C  
divo@MSI:~$ mkdir -p /tmp/monitor_dir  
divo@MSI:~$ ./file_monitor  
Masukkan NIM: 2410651059  
Masukkan path direktori untuk dipantau: /tmp/monitor_dir  
inotify_add_watch: No such file or directory  
divo@MSI:~$ mkdir -p /tmp/monitor_dir  
divo@MSI:~$ ./file_monitor  
Masukkan NIM: 2410651059  
Masukkan path direktori untuk dipantau: /tmp/monitor_dir  
Monitoring directory '/tmp/monitor_dir'...  
Tekan Ctrl+C atau kirim SIGTERM untuk berhenti.  
^C  
divo@MSI:~$ nano myshell.c  
divo@MSI:~$ gcc -o myshell myshell.c  
divo@MSI:~$ ./myshell
```

Tugas 3: Custom Shell

```
divo@MSI:~$ ./file_monitor
Masukkan NIM: 2410651059
Masukkan path direktori untuk dipantau: /tmp/monitor_dir
Monitoring directory '/tmp/monitor_dir'...
Tekan Ctrl+C atau kirim SIGTERM untuk berhenti.
^C
divo@MSI:~$ nano myshell.c
divo@MSI:~$ gcc -o myshell myshell.c
divo@MSI:~$ ./myshell
myshell> pwd
/home/divo
myshell> cd
myshell> cd /tmp
myshell> echo hello nigga
hello nigga
myshell> history
History (last 5 commands):
1: pwd
2: cd
3: cd /tmp
4: echo hello nigga
5: history
myshell> exit
divo@MSI:~$ ./myshell
myshell> sleep 5 &
myshell> sleep 5 &
myshell> ls | grep txt
monitor_log.txt
syscall_test.txt
test_file.txt
myshell> echo hello > file.txt
hello > file.txt
myshell> echo world >> file.txt
world >> file.txt
myshell> cat < file.txt
^C
```

Tugas 4: System Information Tool

```
divo@MSI:~$ nano system_info.c
divo@MSI:~$ gcc -o system_info system_info.c
divo@MSI:~$ ./system_info
=== System Information Tool ===

CPU Model: AMD Ryzen 5 7535HS with Radeon Graphics
CPU Frequency: 3293.886 Mhz
Total Cores: 1

Memory Total: 6783 MB
Memory Free: 5971 MB
Memory Used: 1812 MB

Disk Total: 1031018 MB
Disk Used: 1804 MB
Disk Free: 1029218 MB

Running processes:
PID Name
1 systemd
2 init-systemd(ub
6 init
120 cron
177 dbus-daemon
186 systemd-logind
188 x11-prov-service
206 sshd
214 agetty
222 unattended-upgr
219 login
362 systemd
383 (sd-pam)
385 bash
609 polkitd
1923 systemd-journal
1976 systemd-timesys
2010 systemd-sdssd
2043 systemd-resolve
4286 rsyslogd
8123 SessionLeader
8124 Ralua(8125)
8125 bash
8823 system_info

Network Interfaces:
Interface Bytes Rx Bytes Tx
lo 13314 13314
eth0 79276896 2316622

CPU Load Average: 1min 0.00, 5min 0.00, 15min 0.00
divo@MSI:~$ lscpu
Architecture: x86_64
CPU opmode(s): 32-bit, 64-bit
Address sizes: 48 bits physical, 48 bits virtual
Byte Order: Little Endian
CPU(s): 12
On-line CPU(s) list: 0-11
Vendor ID: AuthenticAMD
Model name: AMD Ryzen 5 7535HS with Radeon Graphics
CPU family: 25
Model: 68
Thread(s) per core: 2
Core(s) per socket: 6
Socket(s): 1
Stepping: 1
BogoMIPS: 6087.61
Flags: fpu_vme de pni tsc mtr pae mce cmi apic sep mtrr pge mca
cmov pat pss36 clflush mmi fxsr sse sse2 ht syscall nx mm
x256 f16c_opt pdep16b rdtscp la constant_tsc rep_good nopl
```

```
divo@MSI: -
Thread(s) per core: 2
Core(s) per socket: 4
Socket(s): 1
Stepping: 1
BogoMIPS: 6587.61
Flags: fpu vme de pae tsc mtr pae mce cr8 apic sep mtrr ppe ara
cmov pat pae36 clflush mmi fvar x86 tsc2 ht syscall nx mm
x87 fpu-ops ppioplgh rdtscp lm constant-tsc rep_good nopl
1 tsc_reliable nonstop_tsc cpuid extd_apicid tsc_100nm_fr
writ pni clflushopt shsti fma cldm shwcl shwcl2 mwaitx pprof
t_auz xsave avx f16c rdprand hypervisor lahf_lm cap_legacy
svm cr8_legacy abm ssse3 misalignsse 3dnowprefetch osvw
rnmovt perfctr_core ssbd sha3 sha3b shstlb vmcall fapdas
q_buil avx2 smap bni2 crng invpcid rdtscv adx smap clflush
hugt clwb sha_512 xsaveopt xsavec optctl tssas clzero ss
avx512pr arat spt nrip.save tsc_scale vmcb_clean flushbysa
sid decompdists pausefilter pfthreshold v_vmsave_mload
unip vaas vpcmlqldq rdpid fvar

Virtualization features:
Virtualization: AMD-V
Hypervisor vendor: Microsoft
Virtualization type: Full
Caches (sum of all):
L1d: 192 MiB (6 instances)
L1i: 192 MiB (6 instances)
L2: 3 MiB (6 instances)
L3: 16 MiB (1 instance)
MTRR:
MTRR mode(s): 1
MTRR mask(s): 0-11
Vulnerabilities:
Cacher data sampling: Not affected
CPU multi-bit: Not affected
L1TF: Not affected
MDS: Not affected
Meltdown: Not affected
Mmio stale data: Not affected
Nep file data sampling: Not affected
NtLm!nd: Not affected
Spec rstack overflow: Vulnerable: Safe RET, no microcode
Spec store bypass: Mitigation: Speculative Store Bypass disabled via prctl
Spectre v1: Mitigation: usercopy/swap barriers and __user pointer s
Spectre v2: Mitigation: Bypass; IBPB conditional; IBRS_FW; STIRP
always-on; RSB filling; PDSB-ALDS Not affected; BHI Not
affected
Srbds: Not affected
Tsx async abort: Not affected
divo@MSI:~$ free -m
              total        used        free      shared  buff/cache   available
Mem:           6783        533        5769           3         642        6245
Swap:           2048             0         2048
divo@MSI:~$ df -h
Filesystem      Size  Used Avail Use% Mounted on
none            3.9G  0 3.9G  0% /usr/lib/modules/6.6.87.2-microsoft-standard-WSL2
none            3.9G  4.0M  3.9G  1% /mnt/ssl
drivers         659G  363G  98G  55% /usr/lib/ssl/drivers
/dev/fd0        1007G  1.8G  950G  1% /mnt/ssl
none            3.9G  0 3.9G  0% /usr/lib/ssl/lib
rootfs          3.9G  2.7M  3.9G  1% /init
none            3.9G  552K  3.9G  1% /run
none            3.9G  0 3.9G  0% /run/lock
none            3.9G  0 3.9G  0% /run/shm
none            3.9G  100M  3.9G  1% /mnt/ssl/versions.txt
none            3.9G  100M  3.9G  1% /mnt/ssl/doc
C:\             659G  363G  98G  55% /mnt/c
tmpfs           679M  20K  679M  1% /run/user/1000
divo@MSI:~$ ps -e
PID TTY          TIME CMD
1 ?            00:00:02 systemd
2 hvcd        00:00:00 init-systemd(ub
```

```
divo@MSI: -
Spec rstack overflow: Vulnerable: Safe RET, no microcode
Spec store bypass: Mitigation: Speculative Store Bypass disabled via prctl
Spectre v1: Mitigation: usercopy/swap barriers and __user pointer s
Spectre v2: Mitigation: Bypass; IBPB conditional; IBRS_FW; STIRP
always-on; RSB filling; PDSB-ALDS Not affected; BHI Not
affected
Srbds: Not affected
Tsx async abort: Not affected
divo@MSI:~$ free -m
              total        used        free      shared  buff/cache   available
Mem:           6783        533        5769           3         642        6245
Swap:           2048             0         2048
divo@MSI:~$ df -h
Filesystem      Size  Used Avail Use% Mounted on
none            3.9G  0 3.9G  0% /usr/lib/modules/6.6.87.2-microsoft-standard-WSL2
none            3.9G  4.0M  3.9G  1% /mnt/ssl
drivers         659G  363G  98G  55% /usr/lib/ssl/drivers
/dev/fd0        1007G  1.8G  950G  1% /mnt/ssl
none            3.9G  0 3.9G  0% /usr/lib/ssl/lib
rootfs          3.9G  2.7M  3.9G  1% /init
none            3.9G  552K  3.9G  1% /run
none            3.9G  0 3.9G  0% /run/lock
none            3.9G  0 3.9G  0% /run/shm
none            3.9G  100M  3.9G  1% /mnt/ssl/versions.txt
none            3.9G  100M  3.9G  1% /mnt/ssl/doc
C:\             659G  363G  98G  55% /mnt/c
tmpfs           679M  20K  679M  1% /run/user/1000
divo@MSI:~$ ps -e
PID TTY          TIME CMD
1 ?            00:00:02 systemd
2 hvcd        00:00:00 init-systemd(ub
6 hvcd        00:00:00 init
176 ?         00:00:00 cron
217 ?         00:00:00 dmcc-daemon
186 ?         00:00:00 systemd-logind
186 ?         00:00:00 wsl-pro-service
286 ?         00:00:00 sshd
214 tty1      00:00:00 agetty
222 ?         00:00:00 systemd-upgr
319 pts/1    00:00:00 login
342 ?         00:00:00 systemd
343 ?         00:00:00 (sd-pam)
345 pts/1    00:00:00 bash
649 ?         00:00:00 polkitd
1921 ?        00:00:00 systemd-journal
1970 ?        00:00:00 systemd-timesyn
2039 ?        00:00:00 systemd-udev
2043 ?        00:00:00 systemd-resolve
4246 ?        00:00:00 rsyslogd
5123 ?        00:00:00 SessionLeader
5124 ?        00:00:00 sshd(5123)
5125 pts/0    00:00:00 bash
6028 pts/0    00:00:00 ps
divo@MSI:~$ cat /proc/net/dev
Inter| Receive
bytes packets errors drop fifo from compressed multicast| bytes packets errors drop fifo colls carrier compressed
eth0: 13314 106 0 0 0 0 0 0 0 0 13314 106 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
eth1: 75379664 35071 0 0 0 0 0 0 0 0 999 2316286 31433 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
divo@MSI:~$ cat /proc/net/dev
0.00 0.00 0.00 1/238 6438
divo@MSI:~$
```